

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A Data Interpretation

Code of Conduct:

I M. Phanith Chandu (192425238) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Phanith chandu

Q. No	Understanding of Data (25%)	Analysiss (25%)	Application of Concepts (20%)	Conclusion (20%)	Clarity (10%)	Total Score (100%)
1						
2						
3						
4						
5						
Total Marks (Out of 100)				Total Marks (Out of 50)		

Annexure A

Data Interpretation

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method Normal Occlusion Low Light

Viola-Jones 80% 60% 55%

YOLO 90% 75% 70%

Deep Learning 95% 85% 80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

2. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions
- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

3. Background Modeling Strategy (GMM)

A GMM-based system shows:

- High false positives in dynamic background
- Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

4. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation

- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

5. Object Detection under Scale Variation

Results:

Object

Size

Accura

cy

Large

94%

Medium 80%

Small 55%

Compare detection performance and evaluate system limitations. Justify improvements.

1. Detection Performance Comparison

Given data:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Precision and Recall Precision = $TP / (TP + FP)$ Recall = $TP / (TP + FN)$ Model A

(Viola-Jones):

- Precision = $80 / (80 + 20) = 80 / 100 = 0.80 = 80\%$.
- Recall = $80 / (80 + 30) = 80 / 110 \approx 72.73\%$.

Model B (YOLO):

- Precision = $90 / (90 + 15) = 90 / 105 \approx 85.71\%$.
- Recall = $90 / (90 + 20) = 90 / 110 \approx 81.82\%$.

(b) Comparison and Evaluation

- YOLO has higher precision: 85.71% compared with 80% for Viola-Jones.
- YOLO has higher recall: 81.82% compared with 72.73% for Viola-Jones.
- YOLO produces fewer false positives (15 vs 20) and fewer false negatives (20 vs 30).
- Higher precision means YOLO's positive detections are more likely to be correct.
- Higher recall means YOLO detects a larger proportion of the actual objects.
- Therefore, YOLO performs better on both important detection measures.

Critical Interpretation

YOLO is the stronger model because it improves both precision and recall rather than improving only one measure. The lower false-positive and false-negative counts provide direct evidence for this decision. For an object-detection application where both reliable detections and fewer missed objects matter, YOLO is preferable.

2. Motion Estimation Error Analysis

Given displacement errors:

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Average Error

Average error = Sum of frame errors / Number of frames

- Method A = $(2 + 3 + 2) / 3 = 7 / 3 \approx 2.33$ pixels.
- Method B = $(3 + 5 + 4) / 3 = 12 / 3 = 4$ pixels.

(b) Comparison and Evaluation

- Method A has an average error of approximately 2.33 pixels.
- Method B has an average error of 4 pixels.
- Method A therefore has approximately 1.67 pixels less average error.
- Method A also has lower error in every individual frame: 2 vs 3, 3 vs 5, and 2 vs 4.

Critical Interpretation

Method A is more accurate because its average error is substantially lower and it performs better in every listed frame. Based strictly on the supplied error data, Method A should be selected when accurate displacement estimation is the main objective.

3. Optical Flow Magnitude Comparison

Given optical flow magnitudes:

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Average Motion Magnitude

Average magnitude = Sum of magnitudes / Number of regions

- Method A = $(5 + 6 + 5) / 3 = 16 / 3 \approx 5.33$ pixels/frame.
- Method B = $(8 + 10 + 9) / 3 = 27 / 3 = 9$ pixels/frame.

(b) Comparison and Evaluation

- Method A has an average motion magnitude of approximately 5.33 pixels/frame.
- Method B has an average of 9 pixels/frame.
- Method B captures approximately 3.67 pixels/frame more average motion magnitude.
- Method B is higher in every region: 8 vs 5 in R1, 10 vs 6 in R2, and 9 vs 5 in R3.
- Therefore, Method B captures stronger motion according to the supplied magnitude

values.

Critical Interpretation

Method B is the stronger method for capturing larger motion magnitude in the given regions. However, a higher magnitude does not automatically mean higher accuracy; it only shows that the measured motion is stronger according to the provided data. The conclusion is therefore specifically about motion magnitude, not overall motion-estimation quality.

4. Pattern Recognition Accuracy Comparison

Given data:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) *Accuracy Calculation*

Accuracy = Correct Predictions / Total Samples × 100

- Traditional = $85 / 100 \times 100 = 85\%$.
- Deep Learning = $92 / 100 \times 100 = 92\%$.

(b) *Comparison and Evaluation*

- Deep Learning has 92% accuracy, while Traditional recognition has 85%.
- Deep Learning is 7 percentage points more accurate.
- Out of 100 samples, Deep Learning correctly recognizes 7 more samples.
- Both systems are evaluated on the same number of samples, so the comparison is direct.
- Based on the supplied accuracy data, Deep Learning performs better.

Critical Interpretation

Deep Learning is preferable when recognition accuracy is the main requirement. The result shows a clear performance advantage of 7 percentage points. The supplied data does not provide processing time, computational cost, or training requirements, so those factors cannot be used to claim that Deep Learning is better in every practical situation.

5. Detection Accuracy Comparison

Given data:

Model	Correct	Total
A	180	200
B	170	200

(a) Accuracy Calculation

Accuracy = Correct Predictions / Total Samples $\times$ 100

- Model A = $180 / 200 \times 100 = 90\%$.
- Model B = $170 / 200 \times 100 = 85\%$.

(b) Comparison and Evaluation

- Model A has 90% accuracy, while Model B has 85%.
- Model A is 5 percentage points more accurate.
- Model A correctly detects 10 more samples than Model B out of the same 200 samples.
- Because both models use the same total number of samples, the accuracy comparison is straightforward.
- Model A performs better according to the supplied accuracy measure.

Critical Interpretation

Model A should be selected if detection accuracy is the primary evaluation criterion. However, the supplied data contains only correct and total counts, so other factors such as latency, computational cost, robustness, and false-positive behavior cannot be evaluated from this dataset.